

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	31 January 3035
Team ID	
Project Name	
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: HouseRent – Online House Rental and Property Booking System

Reference: <https://www.geeksforgeeks.org/system-design/airbnb-system-design/>

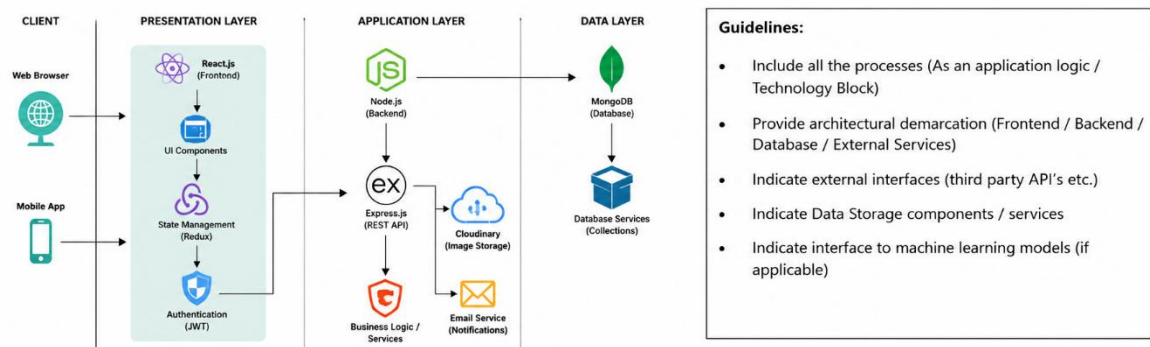


Table-1 : Components & Technologies: (HouseRent Project)

S.No	Component	Description	Technology
1.	User Interface	How users interact with the application e.g. Property listings, forms, dashboards.	HTML, CSS, JavaScript, React.js, Bootstrap / Tailwind CSS
2.	Application Logic - 1 (Backend Services)	Core business logic, APIs, request handling, authentication, authorization.	Node.js, Express.js (JavaScript)
3.	Application Logic - 2 (Services)	Services for property management, booking management, user management, notifications.	Node.js, Express.js
4.	Application Logic - 3 (Background Jobs)	Background jobs for email notifications, scheduled tasks, report generation.	Node-cron, Bull Queue (Redis)
5.	Database	Data storage for users, properties, bookings, payments, reviews, etc.	MongoDB (NoSQL)
6.	Cloud Database	Managed database service on cloud.	MongoDB Atlas
7.	File Storage	Storage for property images, documents, profile pictures.	Amazon S3 / Cloudinary
8.	External API - 1	Payment processing API integration.	Razorpay / Stripe API
9.	External API - 2	Email service for sending notifications.	SendGrid / Mailtrap API
10.	Machine Learning Model	Recommendation engine for suggested properties (if applicable).	Scikit-learn / TensorFlow (ML Model)
11.	Infrastructure (Server / Cloud)	Application deployment on cloud platform. Local Server Configuration : Nginx / PM2 / Docker Cloud Server Configuration : AWS / Render / Vercel / DigitalOcean	AWS EC2, Docker, Nginx, PM2, GitHub Actions (CI/CD)

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	List the open-source frameworks used in the application.	React.js, Node.js, Express.js, MongoDB, Redux Toolkit, Tailwind CSS
2.	Security Implementations	List all the security / access controls implemented, use of firewalls etc.	JWT Authentication, bcrypt (Password Hashing), Role-Based Access Control (RBAC), Input Validation, Helmet.js, CORS, Rate Limiting, HTTPS (SSL/TLS)

S.No	Characteristics	Description	Technology
3.	Scalable Architecture	Justify the scalability of architecture (3 – tier, Micro-services)	3-Tier Architecture (Presentation, Application, Data), RESTful Microservices, Stateless APIs, Horizontal Scaling with Load Balancer
4.	Availability	Justify the availability of application (e.g. use of load balancers, distributed servers etc.)	AWS EC2 Auto Scaling, Application Load Balancer, Multi-AZ Deployment, Database Replication (MongoDB Atlas), 99.9% Uptime
5.	Performance	Design consideration for the performance of the application (number of requests per sec, use of Cache, use of CDN's) etc.	Redis Caching, CDN (CloudFront), Gzip Compression, Database Indexing, Optimized Queries, Response Time < 200ms, Supports 1000+ requests per second

References:

<https://www.mongodb.com/mern-stack>

<https://aws.amazon.com/architecture/microservices/>

<https://react.dev/learn>